



VIGNAN'S

FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH

(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

Module Bank

Module 2

Academic Year 2025-26

Staff Name: Dr. Sreekar Guddeti

Program Name: B. Tech

Branch: AI/ML

Year: 1

Semester: 1

Course: EP

Code: 25PY101

Section number: 28, 34

Instructions:

1. Answer all questions.
2. BT stands for Blooms Taxonomy.

PART-A

1. Quantum Free Electron Theory (QFET)

[10 M]

- a) Compare and contrast the postulates of classical free electron theory with that of quantum free electron theory. [BT 1][6 M]
- b) What are the merits of quantum free electron theory? [BT 1][2 M]
- c) What are the demerits of quantum free electron theory? [BT 1][2 M]

2. Fermi-Dirac Distribution

[10 M]

- a) Plot the probability of occupancy of energy level in the case of Maxwell-Boltzmann distribution and Fermi-Dirac distribution at
 - i. $T = 0K$,
 - ii. $T \neq 0K$, and
 - iii. $T = \infty K$ [BT 2][2 M]
- b) Five free electrons exist in a three-dimensional infinite potential well with all three widths equal to $a = 12 \text{ \AA}$.
 - i. Determine the Fermi energy level at $T = 0K$.
 - ii. Repeat part (i) for 13 electrons. [BT 3][4 M]
- c) Fermi level of a metal is 6.25 eV and electrons in this material follow the Fermi-Dirac distribution. Calculate the temperature at which there is 1% probability that a state 0.30 eV below the Fermi energy level will **not** contain an electron. [BT 3][4 M]

3. Electronic specific heat of solids

[10 M]

- a) Define specific heat of a material. What is its SI unit? [BT 1][2 M]
- b) Specific heat of mercury ($^{200}_{80}\text{Hg}$) is $0.14 \text{ J g}^{-1}\text{K}^{-1}$. Report its specific heat in SI units. [BT 1][2 M]
- c) Derive the expression for specific heat capacity from classical free electron theory and quantum free electron theory. [BT 4][6 M]

4. Density of states

[10 M]

- a) What is the meaning of density of states function? What is its SI unit? [BT 1][2 M]
- b) If the density of states of a material is a constant $Z(E) = k$ and the electron density is n ,
 - i. what is the Fermi level at $T = 0K$?
 - ii. What is the Fermi level at $T \neq 0K$? [BT 3][4 M]
- c) Sketch the density of states function
 - i. for the conduction band of an intrinsic semiconductor.
 - ii. Repeat part (i) for the valence band. [BT 2][4 M]

5. E-k diagram

[10 M]

- a) Sketch the E vs k diagram for a free electron. [BT 1][1 M]
- b) Sketch the E vs k diagram for an electron in a periodic potential. [BT 1][1 M]

- c) Consider an intrinsic semiconductor with all electrons occupying states in the lowest valence band at $T = 0K$. Plot the E vs k showing occupancy of levels at both $T = 0K$ and $T \neq 0K$. [BT 2][2 M]
- d) Two possible conduction bands are shown in the E vs k diagram given in Figure. 1(a). State which band will result in the heavier electron effective mass; state why [BT 2][2 M]
- e) Figure. 1(b) shows the parabolic E vs k relationship in the valence band for a hole in two particular semiconductor materials. Determine the effective mass (in units of the free electron mass) of the two holes. [BT 3][4 M]

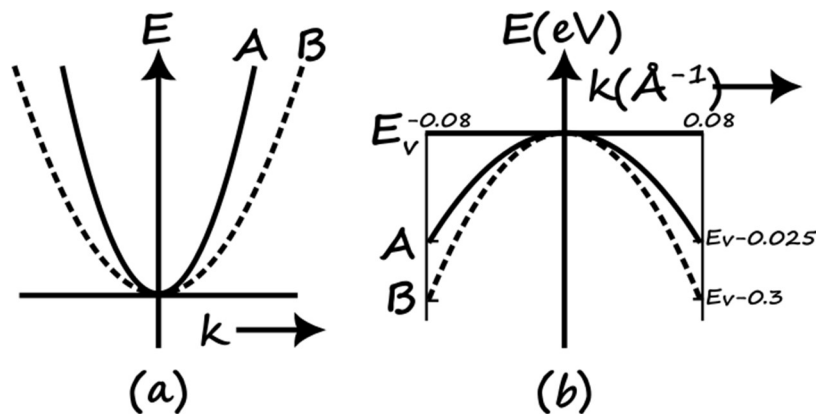


Figure 1: (a) Conduction and (b) valence bands of materials A and B.

6. Fermi level in semiconductors

[10 M]

- a) Sketch the energy band diagram showing the density of states, Fermi-Dirac probability function and areas representing electron and hole concentration for an intrinsic semiconductor at $T \neq 0K$. Assume electron effective mass is equal to hole effective mass. [BT 1, 2][3 M]
- b) Calculate position of Fermi level with respect to center of the bandgap in silicon for
- $T = 0 K$, and
 - $T = 300 K$.
- The electron effective mass is $1.08m_0$ and hole effective mass is $0.56m_0$. [BT 2][2 M]
- c) Draw, compare, and contrast the sketches of the energy band diagram for p-type semiconductor at $T = 0K$ and $T = 300K$. [BT 3][2 M]
- d) Silicon atoms, at a concentration of $7 \cdot 10^{15} cm^{-3}$, are added to GaAs. Assume that the silicon atoms act as fully ionized dopant atoms and that 5% of the dopants replace Ga atoms and 95% replace As atoms.
- Determine the donor and acceptor concentrations.
 - Is the material n type or p type?
- [BT 2][3 M]

7. p-n junction diode

[10 M]

- a) Write the equation and sketch the $I - V$ characteristics of pn junction diode. [BT 1][2 M]
- b) Define reverse saturation current of pn junction diode. Derive it from the $I - V$ characteristics. [BT 1][2 M]

- c) A silicon diode can be used to measure temperature by operating the diode at a fixed forward-bias current. The forward bias voltage is then a function of temperature. Assume that the reverse saturation current is proportional to $\exp(-E_g/k_B T)$. At $T = 300K$, the diode voltage is found to be $0.60V$. Determine the diode voltage at $T = 310K$. What is the percentage change in voltage? [BT 4][3 M]
- d) The pn junction diode can also be used as a temperature sensor by operating the diode at a fixed reverse-bias voltage. What is the percentage change in current for the same temperature rise? [BT 4][2 M]
- e) Which mode is more sensitive, the constant forward current mode or constant reverse voltage mode? [BT 2][1 M]

8. Solar cell

[10 M]

- a) Sketch the $I - V$ characteristics of pn junction solar cell. [BT 1][1 M]
- b) Define short circuit current and open circuit voltage. [BT 1][2 M]
- c) A silicon solar cell under standard test conditions has short circuit current density of $35 A/cm^2$, and open circuit voltage of $0.60 V$. The fill factor for the material is 0.80 . If the incident power is $100 mW/cm^2$, calculate the efficiency of the cell. [BT 3][4 M]
- d) The energy band gap of a semiconductor is $3 eV$. Is this material useful for solar cell applications? Give reasons. [BT 4][3 M]

9. Direct and indirect band gap semiconductors

[10 M]

- a) Sketch the E vs k diagram for
- a direct band gap semiconductor, and
 - an indirect band gap semiconductor.
- [BT 1][2 M]
- b) Why is direct band gap semiconductor useful for solar cell application? Crystalline silicon is indirect band gap semiconductor. What is the process employed in industry to “convert” it into a direct band gap semiconductor for solar cell application? [BT 2][2 M]
- c) Which material is used in blue LED? What is the bandgap of the material? [BT 3][3 M]
- d) $Al_xGa_{1-x}As$ is a ternary semiconductor and the energy band gap variation as a function of mole fraction x of $AlAs$ is shown in Figure 1. Which composition can be used as material for blue LED application? [BT 4][3 M]

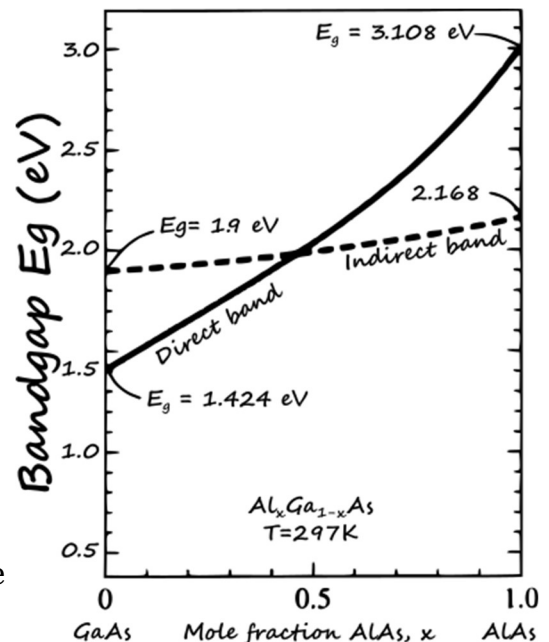


Figure 1. Energy Bandgap

10. LED

[10 M]

- What are the materials used for red, green, and blue LEDs? [BT 1][2 M]
- What are the mechanisms that decrease the efficiency of an LED? [BT 1][3 M]
- The energy band gap variation as a function of mole fraction x of $GaAs_{1-x}P_x$ is shown in Figure 1. What is the range of colours that can be emitted if used for LED application? Report in wavelength. [BT 3][3 M]
- Calculate the critical angle at GaAs - air interface. The refractive index of GaAs is 3.8 at $\lambda = 0.70 \mu m$. [BT 3][2 M]

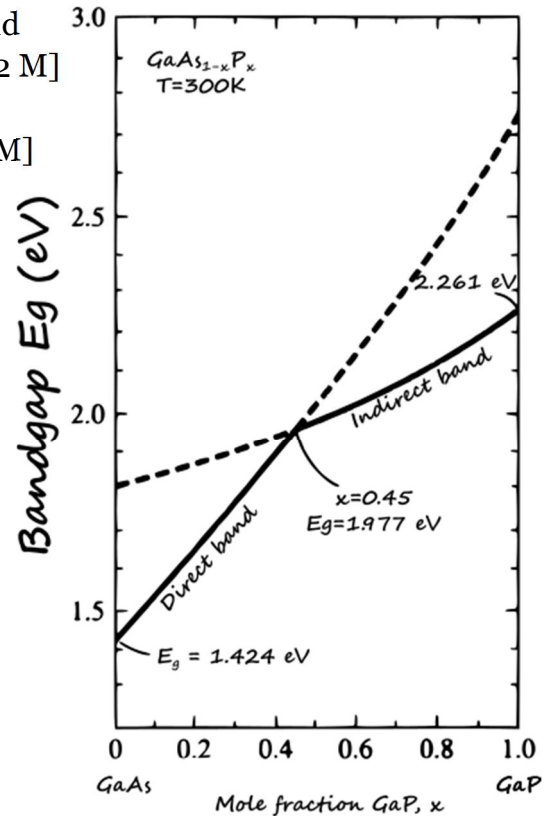


Figure 1. Energy band gap

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